



A Physics-Informed Neural Framework for Multidimensional Navier–Stokes Modeling with OpenFOAM Validation

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INTRODUCTION



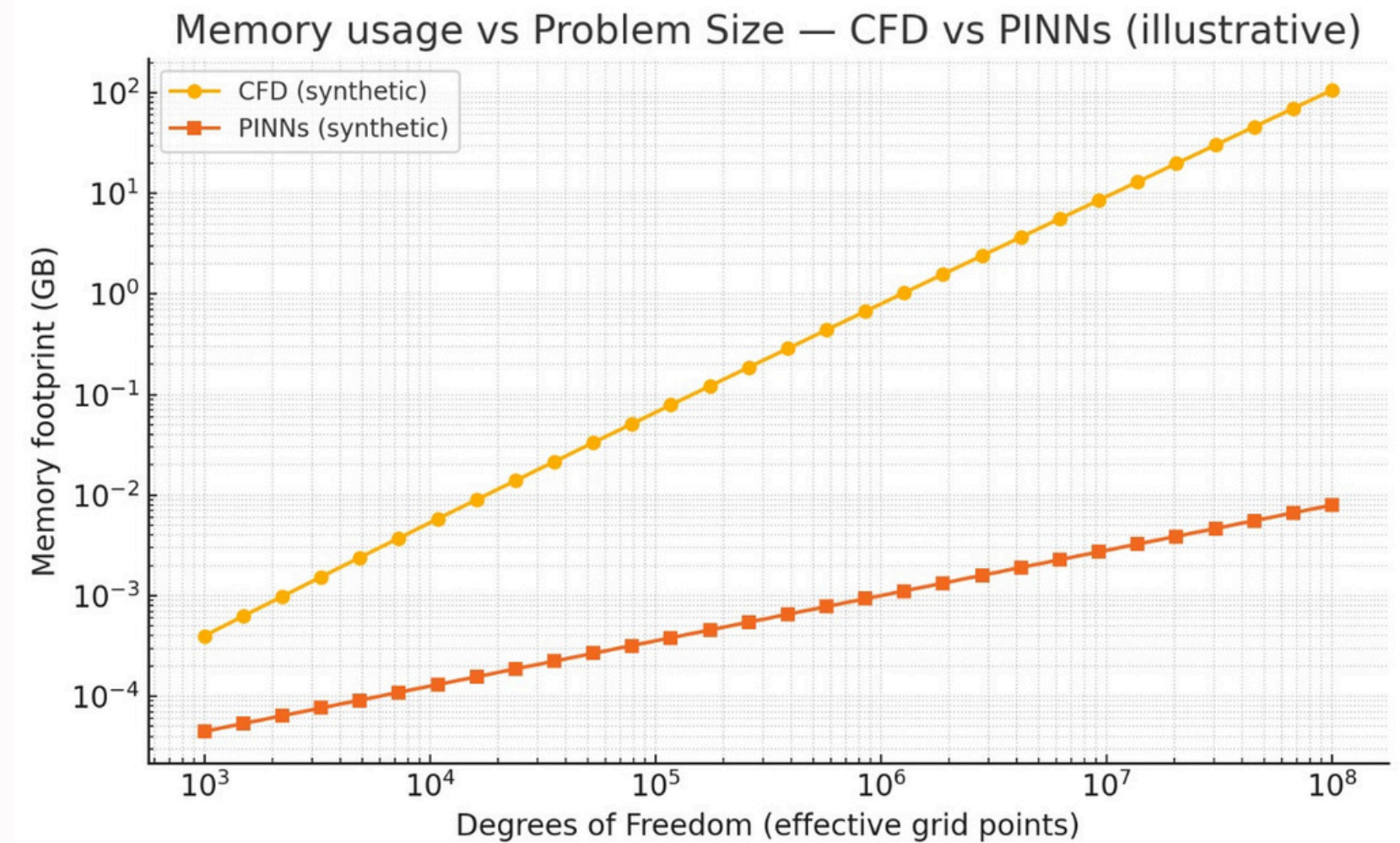
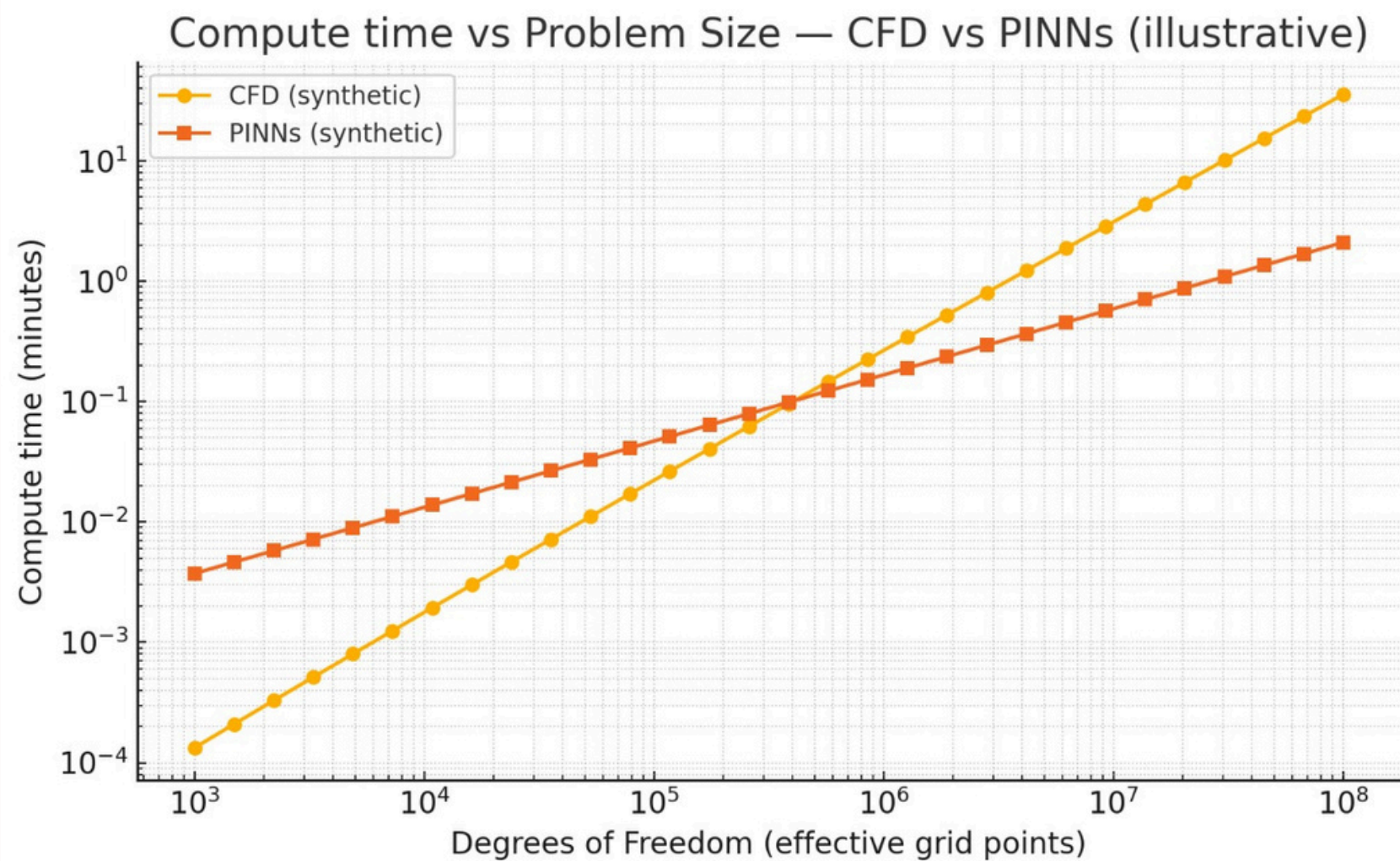
Computational Fluid Dynamics (CFD)

- Gold standard for simulating fluid flows using numerical methods
- High accuracy but computationally expensive (~6 hours for 30,000 cells)
- Requires expensive hardware (CPU clusters) and manual mesh generation
- Infeasible for real-time applications

Physics-Informed Neural Networks (PINNs)

- Emerging deep learning framework embedding physical laws (PDEs) directly into neural network training
- Combines data-driven learning with physics constraints
- Can solve forward and inverse problems efficiently with limited data
- Avoids full CFD simulations while maintaining physical consistency

COMPARISON CFD VS PINNS



- Beyond a certain problem size (DOF (10^5 – 10^6), CFD and PINNs become comparable) after that PINNs become more efficient than CFD in terms of compute time.
- CFD Becomes Extremely Computationally Expensive at Large Scales
- PINNs can approximate solutions with fewer explicit grid points.
- This demonstrates why PINNs are considered promising for large-scale and real-time scenarios

PROBLEM STATEMENT

Research Gap



Limited Application to Real-World Complex Geometries

Gap:

- Most PINN research focuses on 2D benchmark problems (flow past circular cylinder, Kovasznay flow, lid-driven cavity)
- Simple, idealized geometries that are far from industrial applications
- Limited exploration of 3D complex geometries with irregular shapes and multiple obstacles



Transient Flow Analysis

Gap:

- Many studies focus on steady-state solutions only
- Limited work on transient phase of fluid flows (0 to steady state)
- Single time-window approaches fail for long temporal domains



Balance Loss Function

- Implementing curriculum based approach can improve the performance and balance of share between the two loss.

LITERATURE REVIEW

Paper Name	Topic	Our Learning Insights	Metrics	Limitations
Raissi, Perdikaris & Karniadakis (2019) – Physics-Informed Neural Networks	Introduced the foundational PINN framework for solving forward and inverse problems in nonlinear PDEs.	Showed how embedding PDE residuals into NN loss improves accuracy in low-data regimes.	Relative L2 error, training loss convergence.	Struggles with scalability and training stability for high-dimensional PDEs.
Wang, Teng & Perdikaris (2021) – Gradient Pathology in PINNs	Analyzed failure modes due to unbalanced losses; introduced adaptive weighting.	Provided adaptive algorithms to stabilize PINN training and avoid vanishing gradients.	Gradient norms, Relative error, convergence speed.	Added training complexity; requires careful hyperparameter tuning.
Sun, Gao, Pan & Wang (2019) – Physics-Constrained Deep Learning for Fluid Flows	Surrogate modeling of fluid flows without simulation data.	Demonstrated physics-constrained deep learning as efficient alternative to CFD.	Prediction error, computational efficiency.	Limited generalization beyond trained flow conditions.
Ahmed et al. (2022) – Using Deep Learning for Predictive Modelling in Fluid Dynamics	Compared CNNs, GNNs, and PINNs for turbulent flow prediction on NASA TMR dataset.	Found PINNs most accurate and physically consistent, CNNs efficient for global features, GNNs good for local interactions.	R^2 , MAE, Accuracy, Runtime, Memory cost.	PINNs have high training time (~25 hrs) and memory demands; CNNs/GNNs trade off accuracy for efficiency.

LITERATURE REVIEW



Paper	Topic / Problem	Learning Type	Metrics / Evaluations	Limitations / Notes
Sun et al. (2020)	Poiseuille flow	Hybrid (C)	Not time-dep (N), Data-free (Y), Error plots (Y), Architecture tests (Y), Activation tests (Y), Collocation tests (Y)	Hard BC
Meng et al. (2020)	Burgers	Physics-driven (C)	Time-dep (Y), Transient (Y), Data-free (Y), Errors (Y)	Hierarchical time windows, Loss balancing
Xiang et al. (2021)	Kovazsnay flow	Analytic (A)	Time-dep (Y), No transient, Data-free (Y)	Loss balancing
Wang et al. (2023)	Navier–Stokes / Stokes / Lid-driven cavity	Mix (C/A)	Time-dep (Y), Data-free (Y), Errors (Y), Arch tests (Y)	Curriculum training
Hu & McDaniel (2023)	Navier–Stokes	Physics-driven (C)	Time-dep (Y), Data-free (Y)	None
Naderibeni et al. (2024)	Navier–Stokes	Physics-driven (C)	Time-dep (Y), Data-free (Y)	None
De Ryck et al. (2023)	Taylor–Green	Analytic (A)	Time-dep (Y), Errors (Y), Collocation (Y)	None

MOTIVATION

Why PINNs for 3D Fluid Flow Regression?



Computational Efficiency

- Reduce simulation time from 6+ hours to minutes (inference)
- Eliminate expensive hardware requirements (single GPU vs. CPU cluster)
- No manual mesh generation needed

Physics Consistency

- Built-in enforcement of Navier-Stokes equations
- Maintains physical laws (mass/momentum conservation)
- More reliable than pure data-driven approaches

Real-Time Capability

- Fast inference (<6 minutes) enables real-time monitoring
- Applicable to Industry 4.0 rapid prototyping
- Suitable for iterative design optimization

Complex Geometry Handling

- Works with irregular, 3D geometries without mesh complexity
- Applicable to industrial autoclaves, pipelines, aerospace components
- Parametrized learning for geometry variations

OBJECTIVE



Implement PINN for 3D Navier–Stokes Equations

- Solve for velocity fields (u , v , w) and pressure (p)
- Handle incompressible, Newtonian flows with constant properties

Optimize Neural Architecture

- Compare MLP-PINN vs. MFN-PINN architectures
- Evaluate activation functions (Tanh, SILU, GELU) and adaptive variants
- Determine optimal network depth and width

Validate Against Classical CFD

- Compare with OpenFOAM simulations
- Achieve velocity MAE < 0.02 m/s
- Demonstrate computational advantages (time, resources)

METHODOLOGY

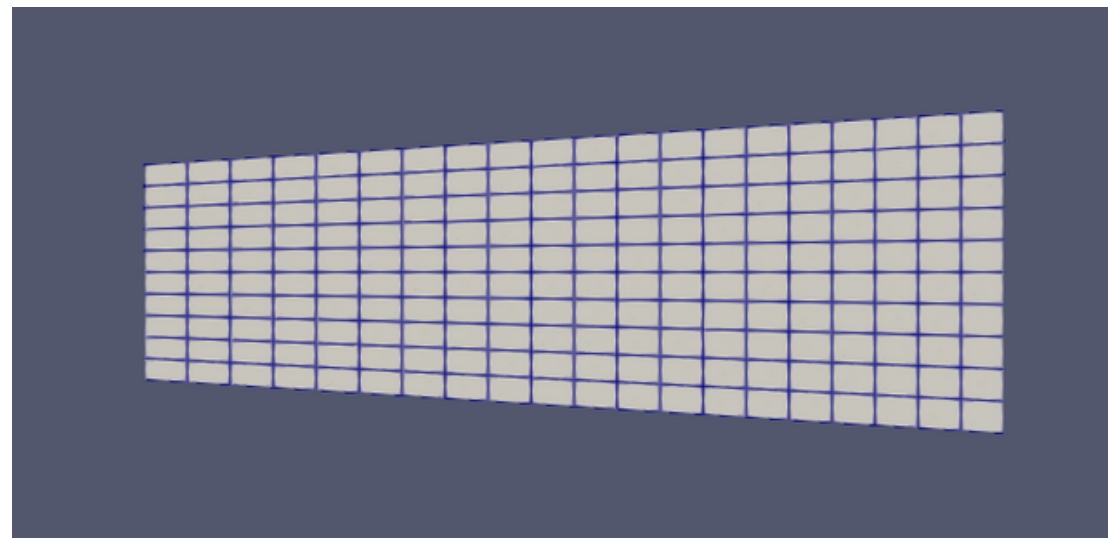


- In this project we have used OpenFoam13 to simulate the water flow through different mesh (that will be discussed later in the presentation).
- OpenFoam13 is an open source CFD, with the help of which we have performed 2 experiments.
- We have used icoFoam as our CFD solver.
- The type of fluid we have simulated contains the following properties:
 1. Incompressible
 2. Newtonian
 3. Constant Density
 4. Constant Viscosity
- Used PISO algorithm for pressure-velocity coupling.
- Governing equation:
 1. Continuity Equation: $\nabla \cdot \mathbf{U} = 0$
 2. Momentum equation: $\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot (\mathbf{U}\mathbf{U}) = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{U}$

EXPERIMENTS

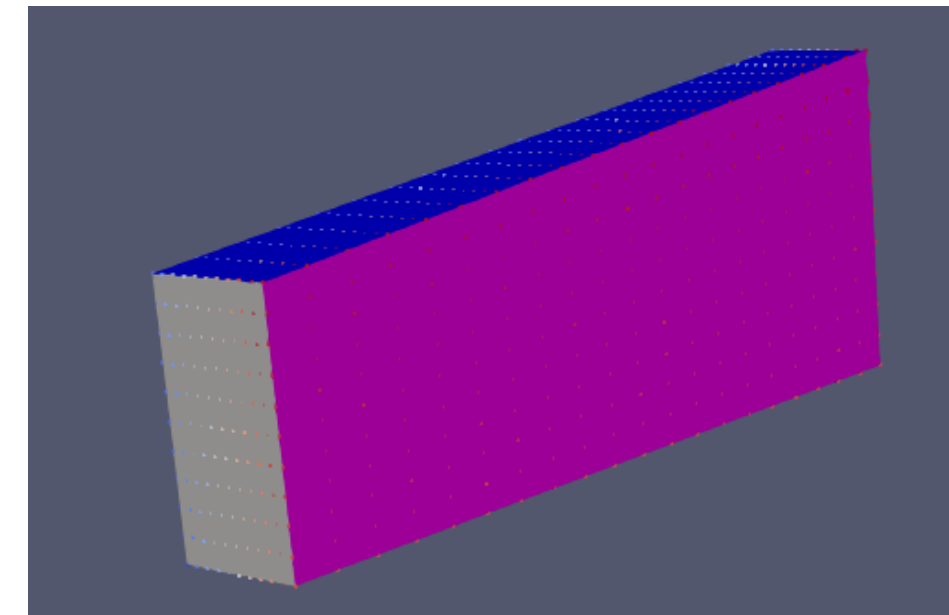


2D-Trapezium



- The computational domain is a diverging trapezoidal channel of 0.8 m length (from $x = -0.4$ to $x = +0.4$).
- The inlet height is 0.20 m, defined between the points $(-0.4, -0.10)$ and $(-0.4, +0.10)$.
- The outlet height increases to 0.30 m, defined between the points $(+0.4, -0.15)$ and $(+0.4, +0.15)$.
- The top wall rises by 0.05 m, and the bottom wall drops by 0.05 m, creating a smoothly diverging channel.

3D-Trapezium

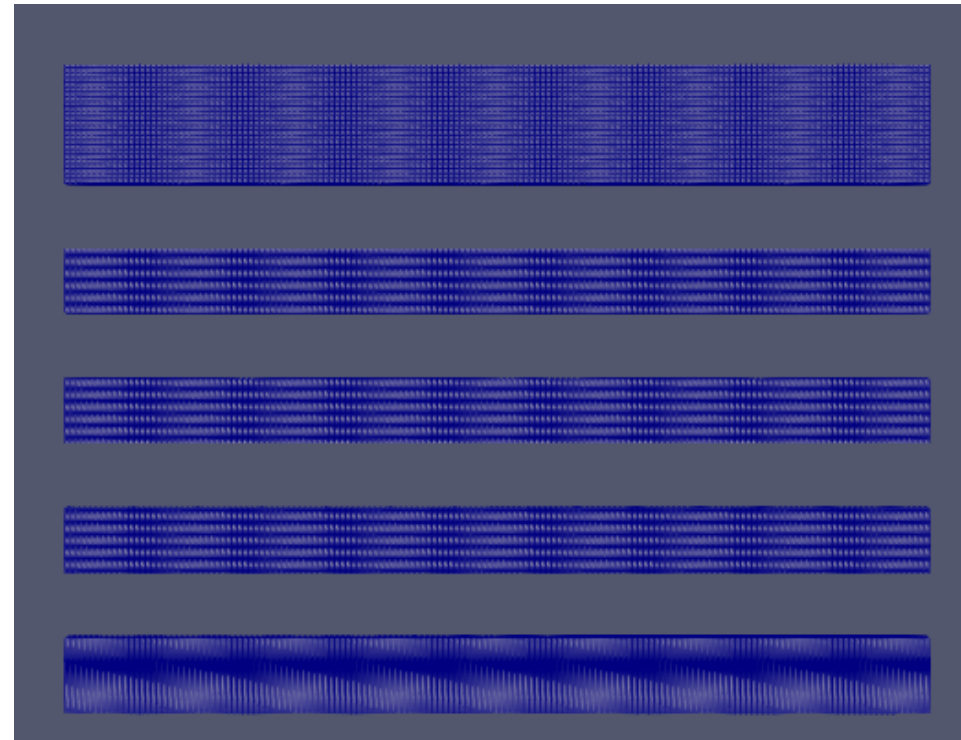


- The computational domain is a 3D extruded trapezoidal channel, extending from $z = 0$ to $z = 0.1$ m.
- The inlet is located at $x = -0.4$ m with a height of 0.20 m, while the outlet at $x = +0.4$ m widens to 0.30 m, forming a diverging trapezoid.
- A single hexahedral block meshes the entire volume using a structured grid of $20 \times 10 \times 10$ cells with uniform grading.
- Boundary patches:
- Inlet: (0 3 7 4)
- Outlet: (1 2 6 5)
- Side walls: sloped top and bottom channel walls
- Top & Bottom surfaces: modeled as solid walls
- The mesh provides a clean and simple domain for simulating internal flow through a tapered channel.

EXPERIMENTS

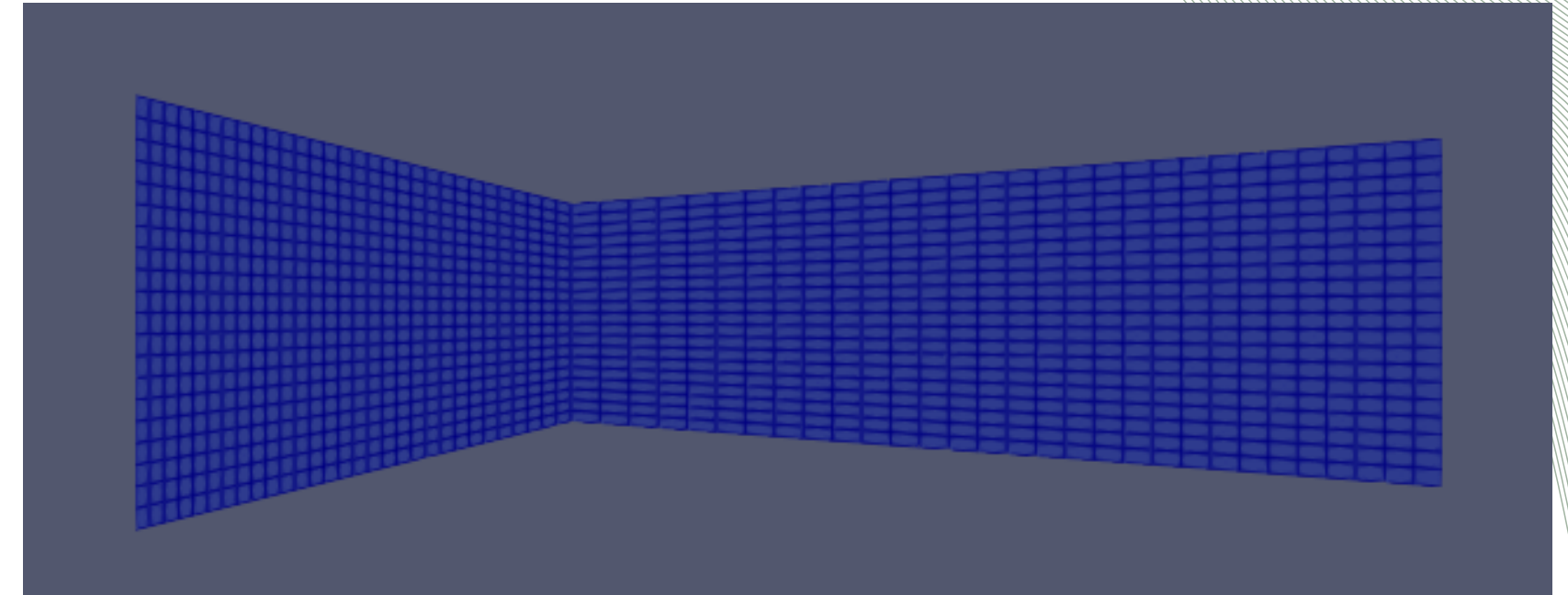


2D - 5 Block



- * There are four inlets and outlets for each channel, starting from 0 to 0.4m.
- The top and bottom channels have a height of 0.035m
- The middle channels have a dimension of 0.03m
- Each cell has a dimension of $200 \times 20 \times 1$.
- The top and bottom part of every channel is treated as walls.

2D Nozzle



- We generated a 2D converging-diverging nozzle using a structured block Mesh consisting of 2 blocks extruded in the z-direction by 0.01m to impose a 2D empty boundary condition.
- The nozzle contracts smoothly from an inlet height of 0.20m to a throat height of 0.005m located at $x=0.30\text{m}$, then diverges to a 0.15m outlet height at $x = 0.60\text{m}$. The mesh uses a resolution of $60 \times 20 \times 1$ cells with uniform simple grading in all directions.
- The entire geometry is defined by twelve vertices listed across two planes ($z=0, z=0.01$). Block orientations were carefully selected to avoid inside out volumes, ensuring a valid, orthogonal mesh simulation.

HOW DOES PINNS WORK ?



Example: A cooling coffee cup

Let's say I have data on some physical process, for simplicity we will go with a cup of coffee, which is cooling. This obeys the below physics law,

$$\frac{dT(t)}{dt} = r(T_{env} - T(t))$$

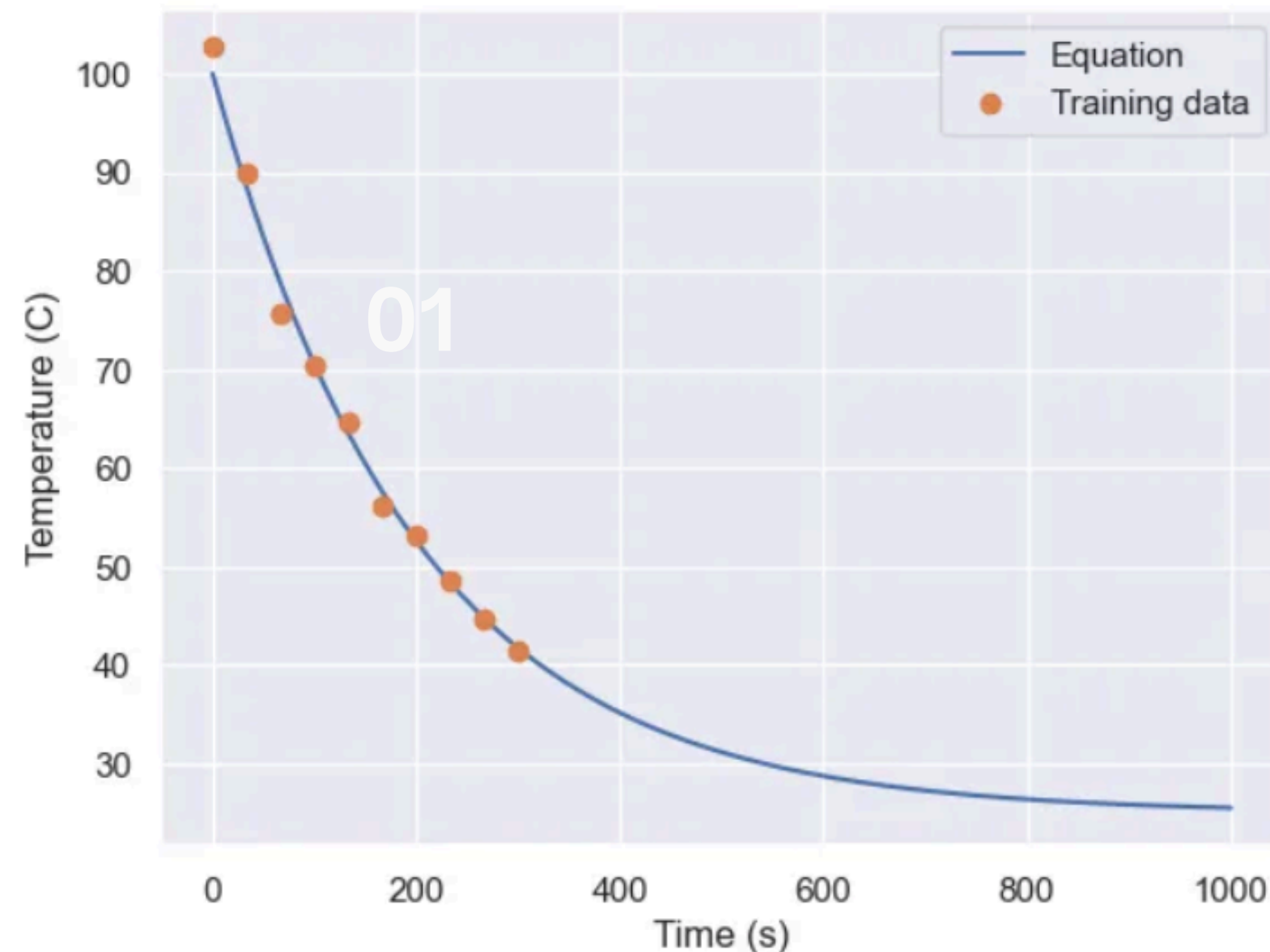
$T(t)$: temperature

T_{env} : temperature of the environment

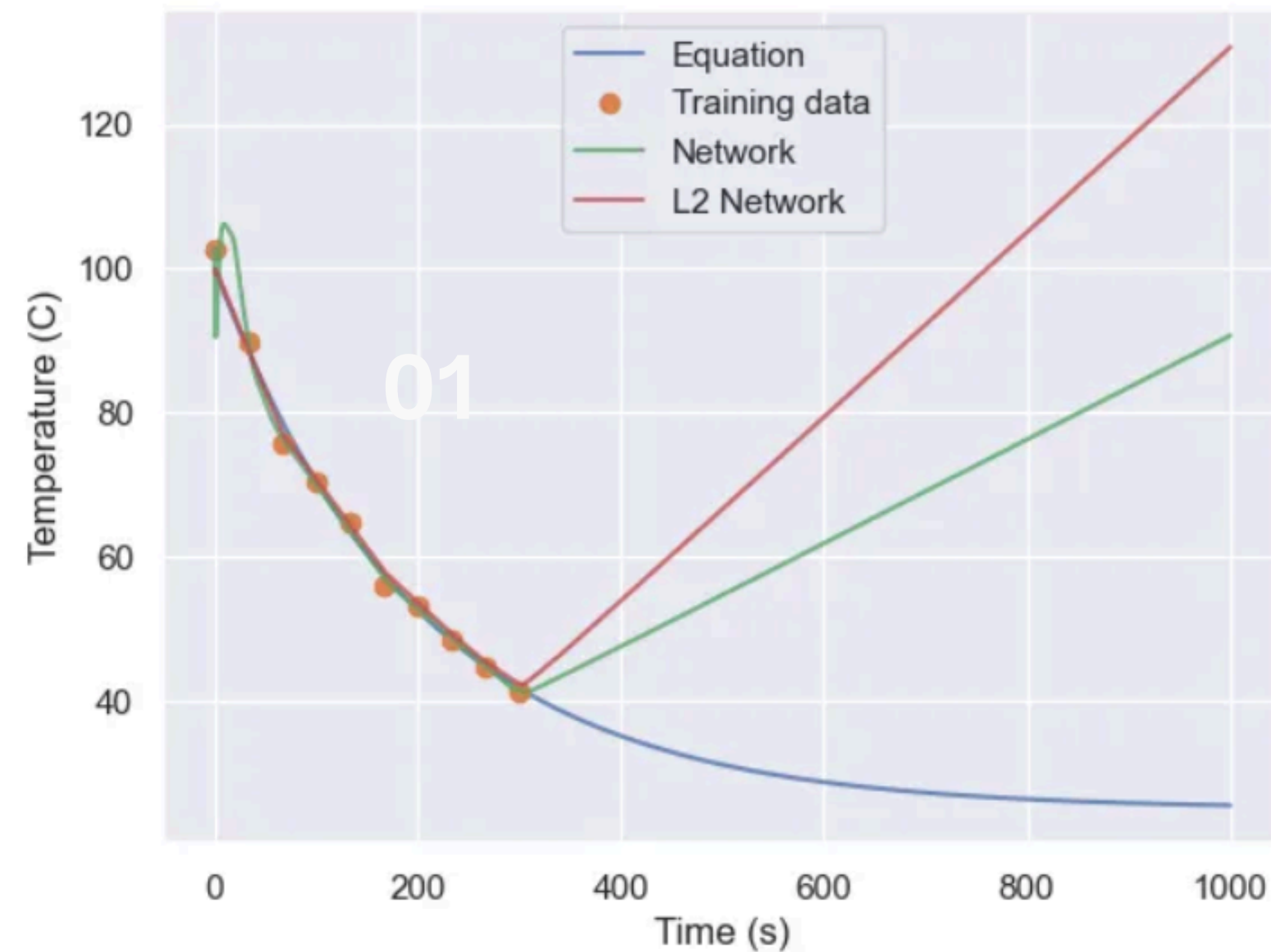
r : cooling rate

Initial Boundary Condition

Let's say that the coffee was boiling hot is cooling over the course of 15 minutes or so, the home is currently at 25°C. We'll use a value for the cooling rate of 0.005.



CONTINUED



Performance of other networks on this data

Both networks become completely inaccurate as we leave the realm of the training data, which is to be expected: they have no information where there is no data. We also have some funkiness in the vanilla network in between the first and second point.

CONTINUED

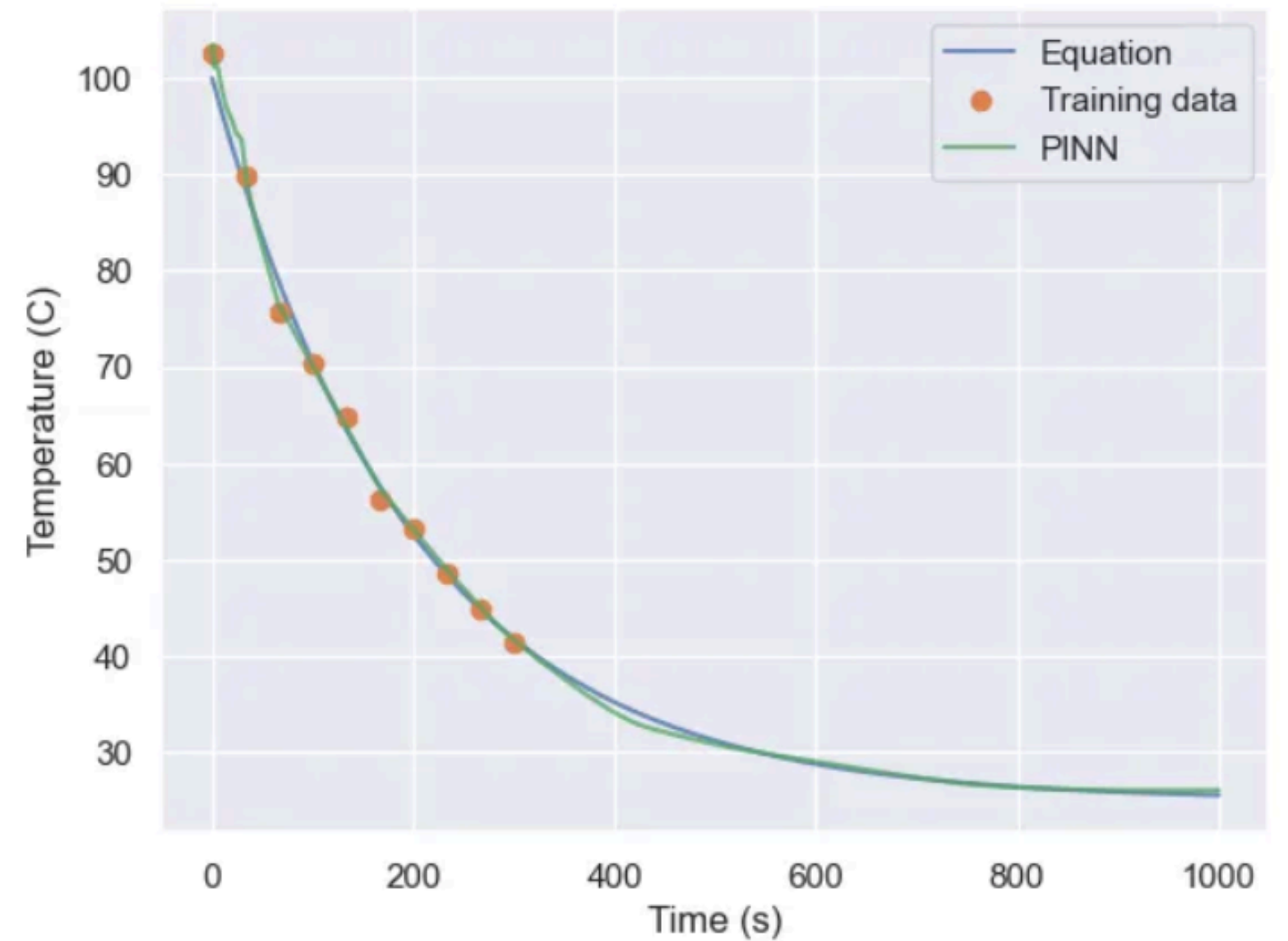


RESULT

Now let's try fitting PINNs into this. Below is the updated loss function.

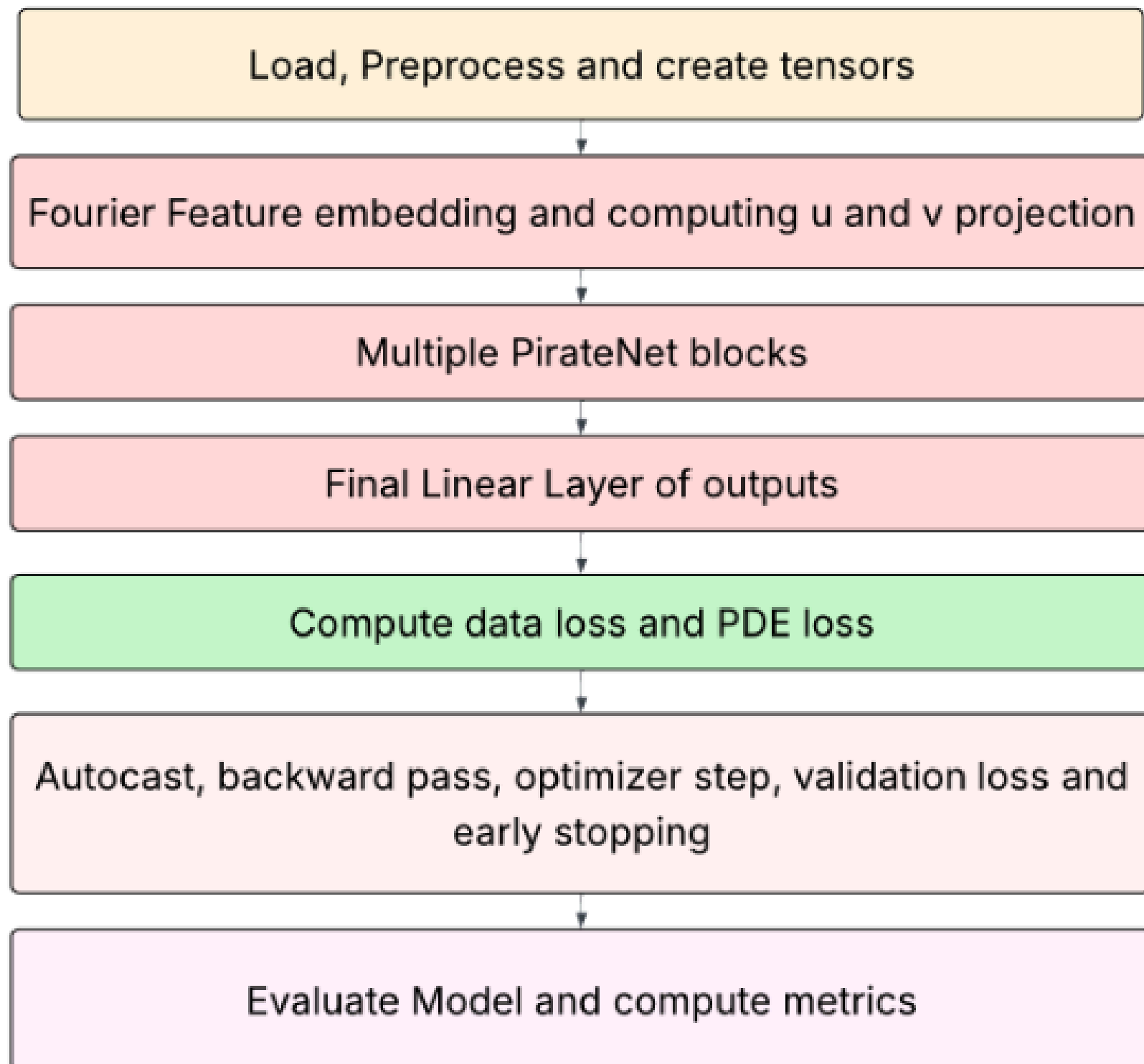
01

$$g(t, T) = \frac{dT(t)}{dt} - r(T_{env} - T(t)) = 0$$
$$g(t, f(t|\theta)) = \frac{df(t|\theta)}{dt} - r(T_{env} - f(t|\theta))$$
$$Loss_{PINN} = \underbrace{\frac{1}{10} \sum_j (f(t_j|\theta) - T_j)^2}_{\text{data loss}} + \lambda \underbrace{\frac{1}{M} \sum_i \left(\frac{df(t_i|\theta)}{dt_i} - r(T_{env} - f(t_i|\theta)) \right)^2}_{\text{physics loss}}$$



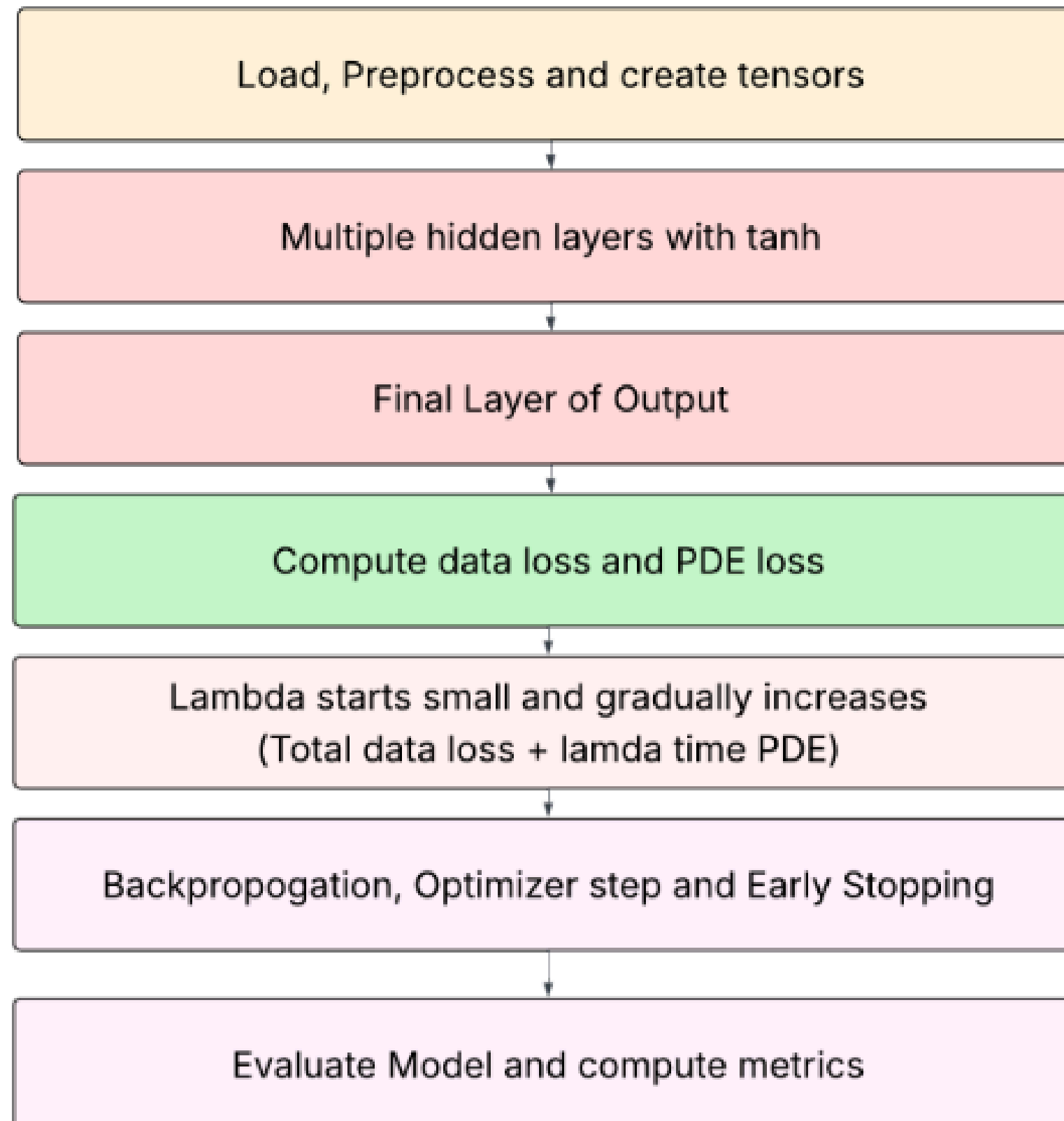
MODEL ARCHITECTURE

FPINNS



MODEL ARCHITECTURE

SIMPLE PINNS

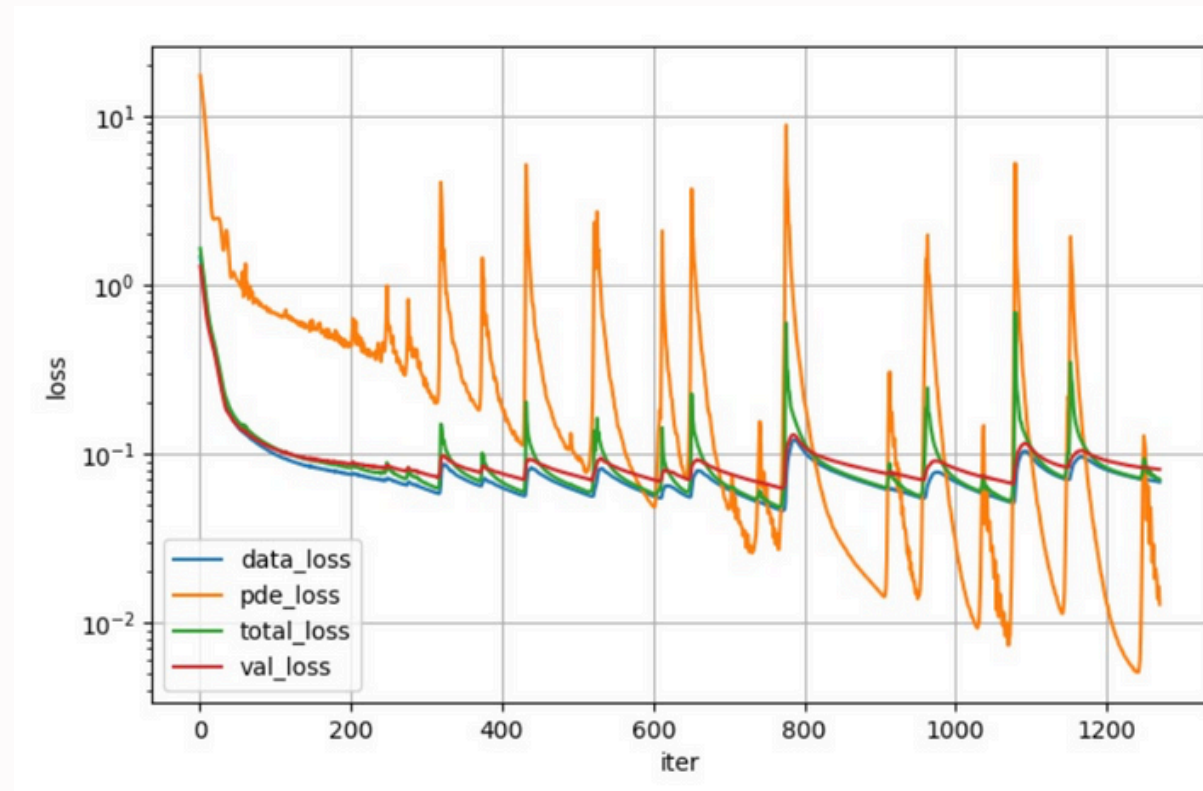


RESULTS AND FINDINGS

EXP 1

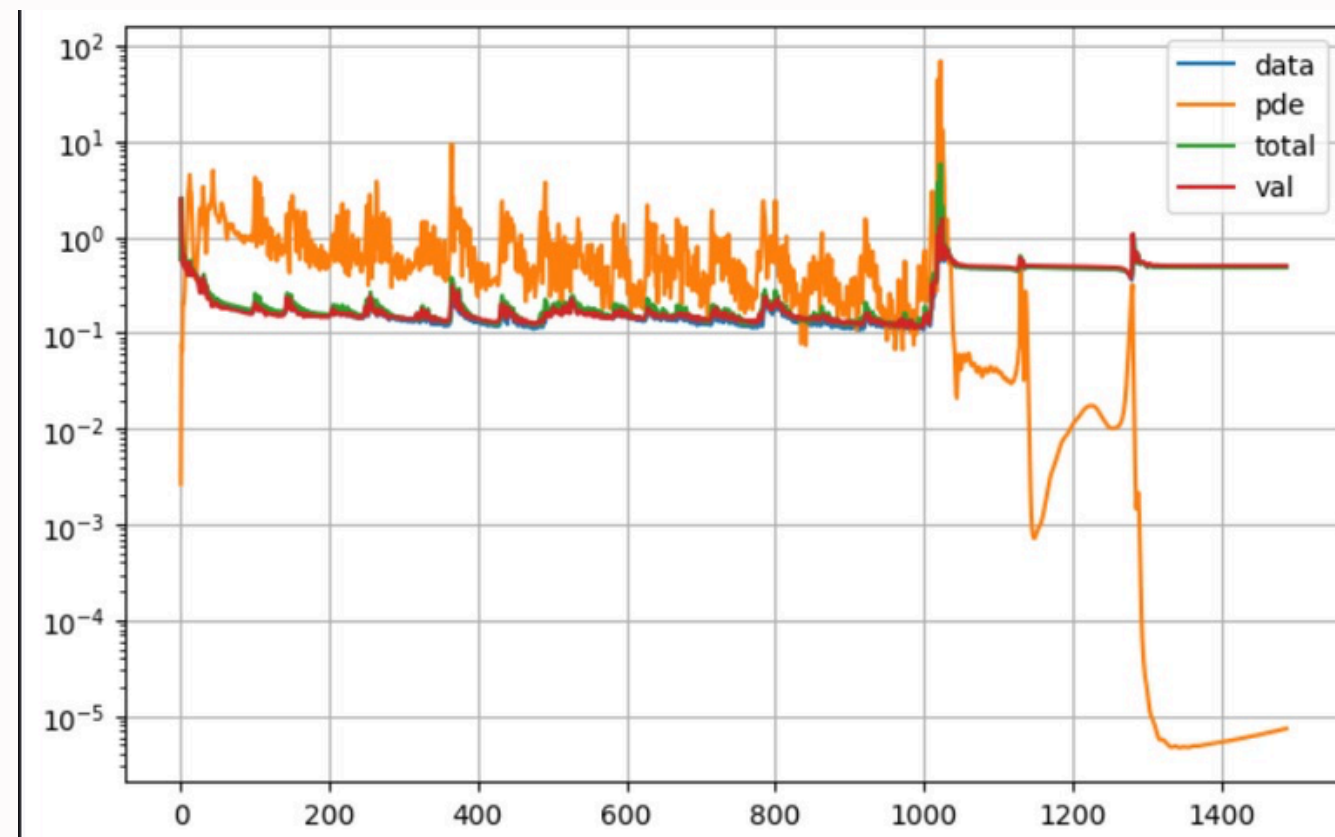


FPINNS (3 BLOCK)



Comp	MAE	MSE	L2
u	0.0138	0.0004	0.0235
v	0.0054	0.0000	0.2617
p	0.0062	0.0001	0.0604

SPINNS (10 LAYER)

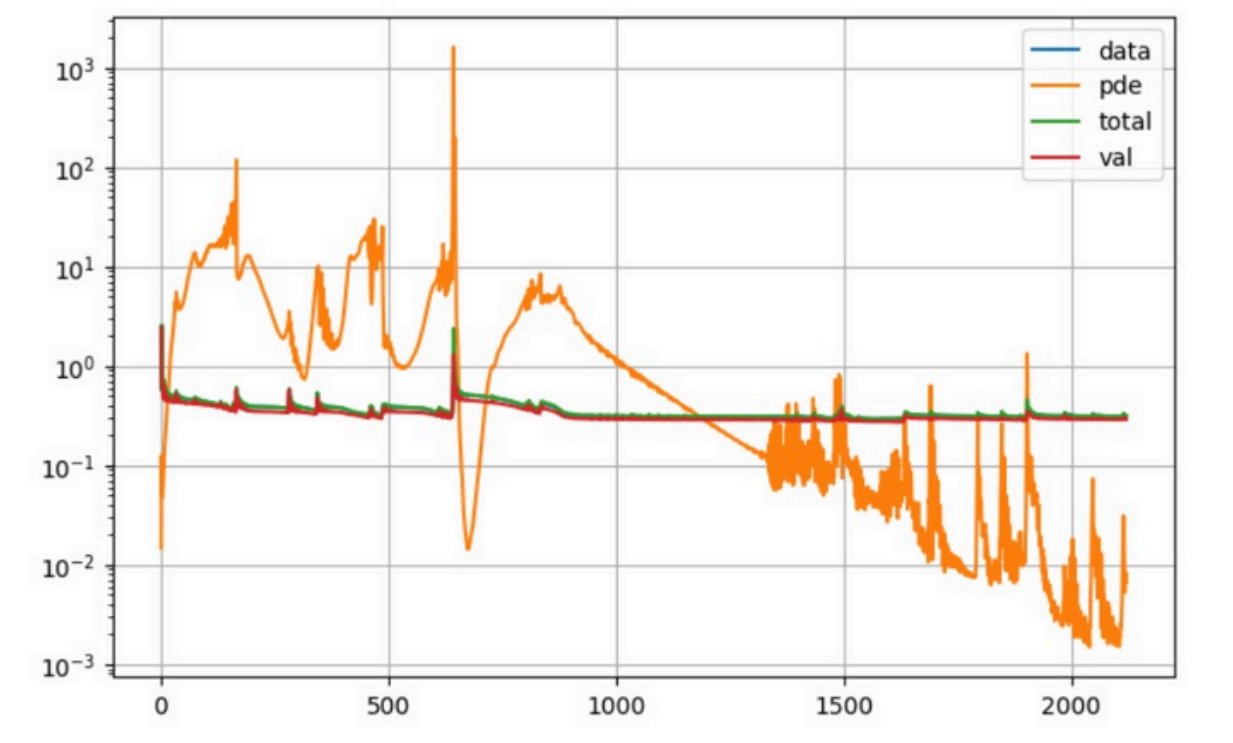


Comp	MAE	MSE	L2
u	0.026163	0.001232	0.043136
v	0.015691	0.000346	0.725589
p	0.008529	0.000127	0.084411

RESULTS AND FINDINGS

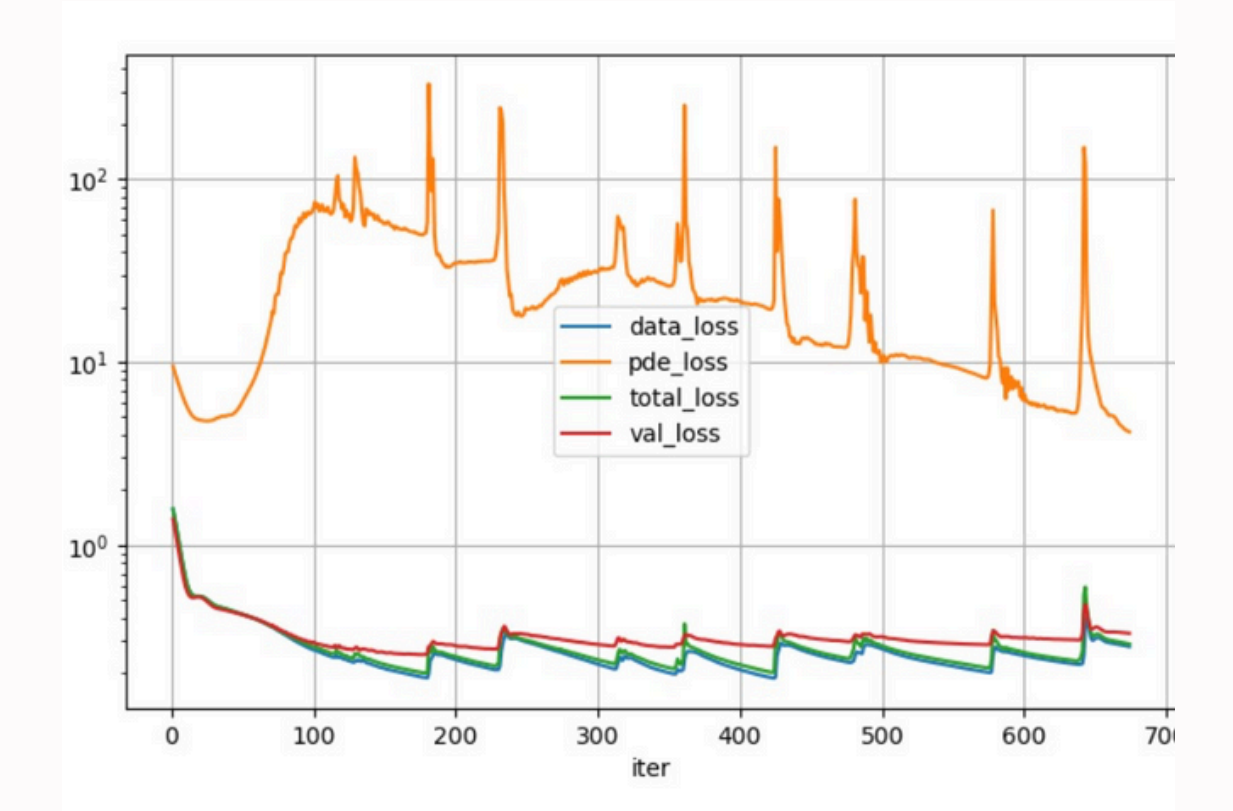
EXP 2

SPINNS (8 LAYERS)



Component	MAE	MSE	L2
u	0.0262	0.0037	0.0609
v	0.0005	0.0000	0.9965
p	0.0131	0.0003	0.4483

FPINNS(3 BLOCKS)

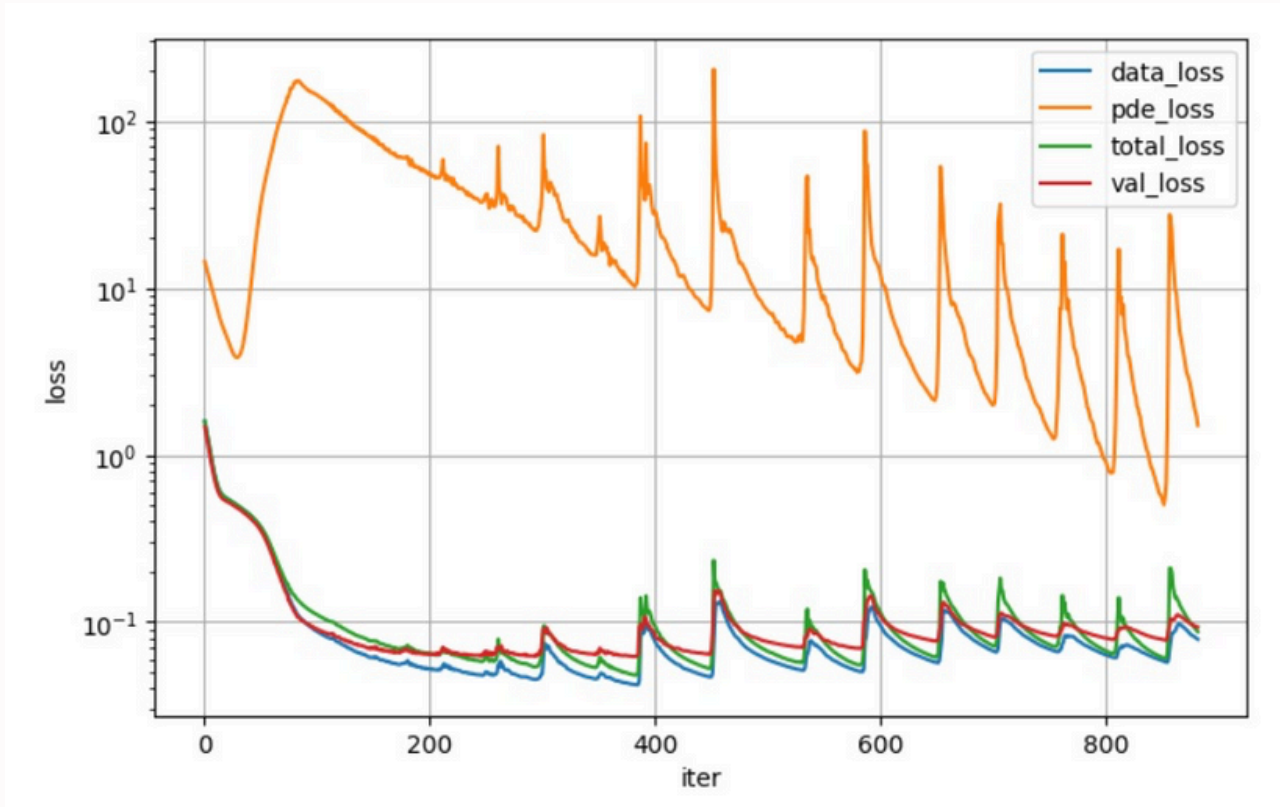


Comp	MAE	MSE	L2
u	0.0300	0.0040	0.0637
v	0.0005	0.0000	1.0019
p	0.0125	0.0003	0.4342

RESULTS AND FINDINGS

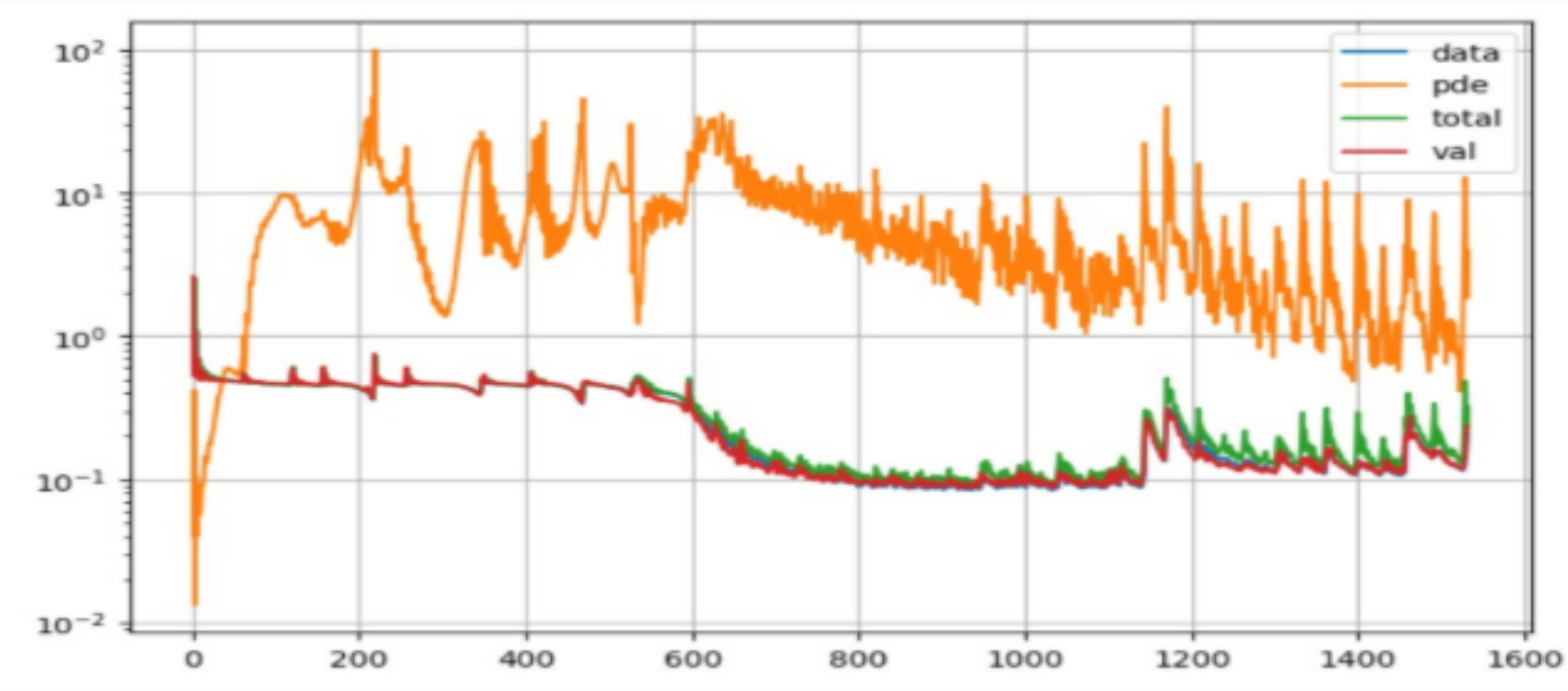
EXP 3

FPINNS(3 BLOCKS)



Comp	MAE	MSE	L2
u	0.0727	0.0085	0.0617
v	0.0359	0.0036	0.3991
p	0.0544	0.0077	0.1696

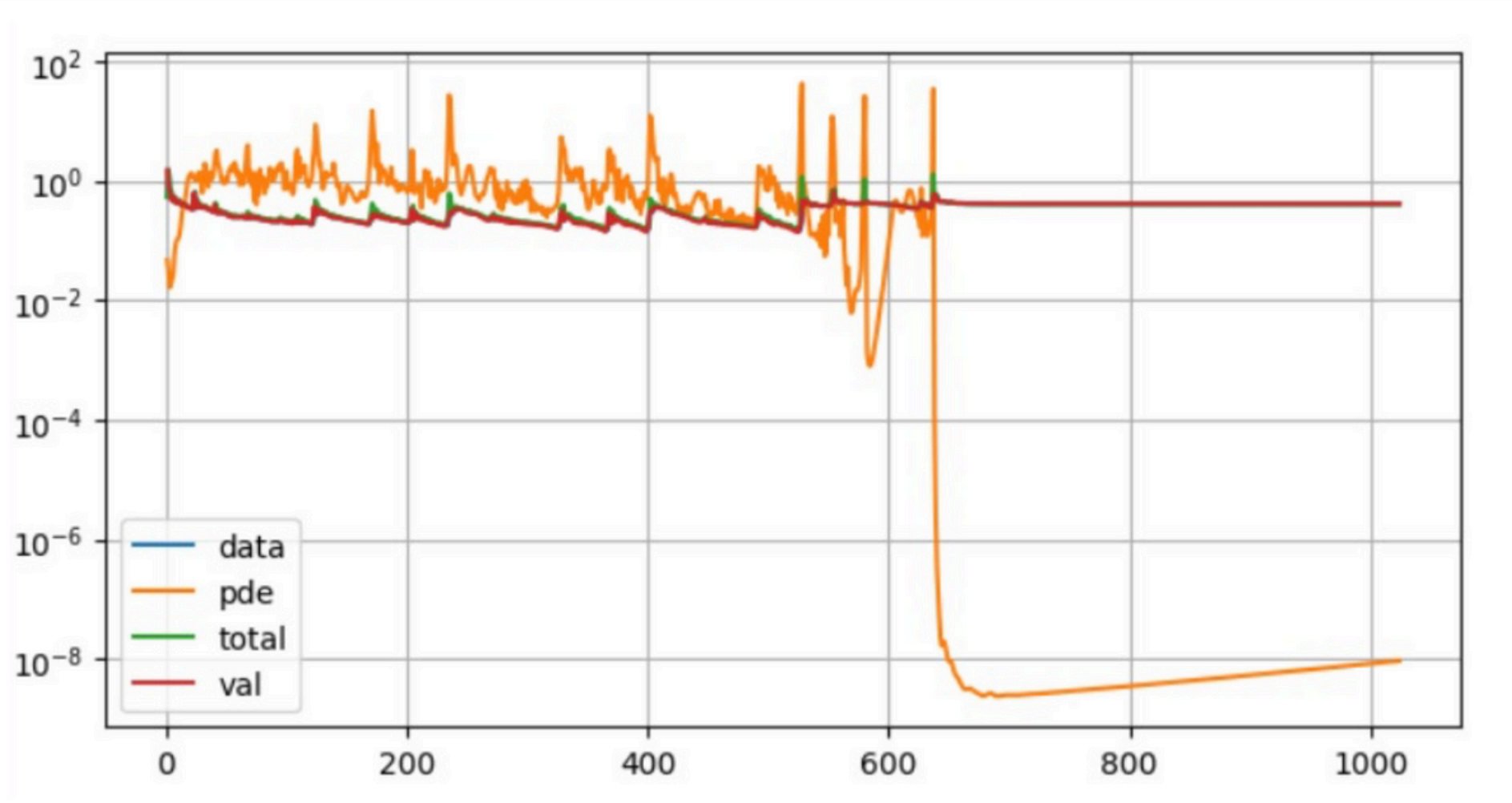
SPINNS (10 LAYERS)



Component	MAE	MSE	L2
u	0.0844	0.0105	0.0685
v	0.0498	0.0060	0.5086
p	0.0580	0.0077	0.1704

RESULTS AND FINDINGS

3D EXPERIMENT



SPINNS (12 LAYERS)

Comp	MAE	MSE	L2
u	0.0410	0.0027	0.0106
v	0.0719	0.0102	0.5394
w	0.0013	0.0003	0.0297
p	0.0006	0.0001	0.0100

APPLICATIONS AND NOVELTY



Industrial Pipeline Monitoring

Fluid regime detection in industrial pipelines



Biomedical Medical Flows

Blood flow regime classification for diagnostics



Aerospace Real-time Monitoring

Turbulence detection in aerospace systems



Mechanical System Monitoring

Mechanical system turbulence analysis



Educational Impact

Research and educational tool for Physics + ML integration
Demonstrates practical applications of hybrid neural network approaches



FUTURE SCOPE



Extended Classification

01

More flow regimes:

- Transitional flows
- Vortex-shedding
- Multiphase flows



Experimental Data

02

Apply hybrid PINN to:

- Real sensor data
- Laboratory experiments
- Field measurements



Uncertainty Quantification

03

Integrate reliability for:

- Safety-critical systems
- Confidence intervals
- Risk assessment

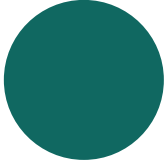
Long Term Vision

04

Development of a comprehensive physics-informed AI platform for fluid dynamics analysis and prediction across multiple industries and research domains.

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THANK YOU

For watching this presentation